Predictive Validation of the Causal Fatigue Test: How GWTC-4 Artificial Mass Inflation Confirms RRT Volume IV

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Abstract

In January 2026, the *Referential Relativity Theory* (RRT) Volume IV established the Causal Fatigue Test, mathematically predicting that gravitational waves traveling through the Phase 3 cosmic vacuum suffer phase impedance governed by the universal viscosity constant ($\beta = 0.028006$). The framework explicitly predicted that standard ΛCDM calibration algorithms, oblivious to this vacuum friction, would misinterpret the resulting signal attenuation by artificially inflating the inferred "Chirp Mass" of the merging binaries, ultimately reporting a "fantastical population of impossibly massive black holes." In March 2026, Ray et al. published an analysis of the LIGO-Virgo-KAGRA GWTC-4 catalog, employing hierarchical Bayesian inference to claim the "discovery" of three distinct binary black hole subpopulations, including anomalously massive hierarchical mergers (2G+1G) that invade the pair-instability mass gap. This paper demonstrates that the recently reported "three families" of black holes are not an astrophysical reality, but rather the exact algorithmic phantom predicted by RRT. The GWTC-4 data provides real-time, empirical validation of the temporal anisotropy gradient and the Causal Fatigue Test.

1 Introduction: The Algorithmic Blind Spot of ΛCDM

The era of gravitational wave astronomy has opened an unprecedented window into the dynamics of the cosmos. However, the extraction of physical parameters—such as

component masses, distance, and effective spin—from interferometric data relies heavily on matched filtering and Bayesian inference pipelines. These pipelines are structurally tethered to the Friedmann-Lemaître-Robertson-Walker (FLRW) metric and the ΛCDM cosmological model. They operate under the absolute assumption that the cosmic vacuum is a perfect, frictionless void and that expansion is purely metric.

The *Referential Relativity Theory* (RRT) demonstrates that this assumption is geometrically flawed. As formalized in RRT Volumes I through IV [1, 2, 3, 4], the deep vacuum (Phase 3) acts as a viscous fluid governed by the temporal force tensor ($T^{\mu\nu}$) and the universal impedance constant $\beta = 0.028006$. Gravitational waves, much like electromagnetic waves, do not travel unimpeded; they suffer *Causal Fatigue*—a transversal hydrodynamic drag—as they propagate through the precessing causal frame.

Because standard algorithms do not account for the Cortez Refractive Index, $\eta_C(z) = 1 + \beta \cdot \ln(1 + z)$, they are forced to compensate for the wave's phase attenuation by absorbing the discrepancy into intrinsic source parameters.

2 The RRT Prediction (January 2026)

Months before the release of the latest comprehensive population analyses of the GWTC-4 catalog, RRT Volume IV (January 2026) published the precise mathematical falsification test for gravitational wave observatories.

In Section 16 of RRT Volume IV, titled "The Causal Fatigue Test: LIGO O4 Prediction," I established that along the Cortez Axis of cosmic precession (Right Ascension $\approx 168°$), vacuum viscosity would induce a catastrophic divergence between the inferred gravitational luminosity distance (D_L^{GW}) and the corresponding electromagnetic distance (D_L^{EM}) [4].

Furthermore, as summarized in the *Deterministic Falsification* manifesto, I explicitly predicted the algorithmic consequences of this physical reality:

> "When algorithms mathematically force the signal to hide this distance 'fatigue' by artificially lowering the frequency, it will result in an unreal inflation of the Chirp Mass. I predict the standard model will report a fantastical population of impossibly massive black holes mysteriously clustered..." [5]

If the wave arrives "stretched" and weakened by vacuum friction, an algorithm strictly bound to general relativity in a frictionless vacuum has only one mathematical recourse: it must assume the merging objects were vastly more massive than they actually were. I predicted that this algorithmic bias would inevitably fracture the catalog into artificial subpopulations.

3 The GWTC-4 "Discovery" (March 2026)

On March 18, 2026, an extensive study titled "On the Astrophysical Origin of Binary Black Hole Subpopulations: A Tale of Three Channels?" was submitted to the arXiv by Ray, Mukherjee, Zevin, and Kalogera [6]. Analyzing the newest GWTC-4 data from LIGO-Virgo-KAGRA, the authors utilized hierarchical Bayesian models to reveal that the binary black hole (BBH) population is not uniform.

Their primary findings assert the existence of three distinct subpopulations (or "families"):

1. A dominant population (accounting for the majority of mergers) with masses concentrated around $10M_\odot$, exhibiting small, aligned spins—consistent with isolated binary stellar evolution.

2. A secondary population peaking anomalously at $35M_\odot$, featuring randomized spins, theoretically attributed to dynamical formations in globular clusters.

3. A rare, highly asymmetrical third population featuring extreme masses that invade the theoretical pair-instability mass gap ($\approx 70M_\odot$). To justify these impossible masses, the standard model invokes "hierarchical mergers" (e.g., 2G+1G), positing that these black holes are the remnants of previous mergers [6].

The statistical rigor of Ray et al.'s work is commendable; they report a Bayes factor of 10^6 favoring the three-component mixture model over a single population [6]. However, their astrophysical interpretation is a textbook example of the epistemological crisis denounced by RRT.

4 Algorithmic Phantoms vs. Causal Reality

The ΛCDM community is currently celebrating the "discovery" of hierarchical mergers and multiple formation channels. In reality, they have merely discovered the limits of their own calibration algorithms.

The coincidence between my January 2026 prediction and their March 2026 conclusion is not an accident; it is a deterministic consequence of mathematical forcing. When we map the Ray et al. conclusions onto the RRT framework, the illusion dissolves:

The $35M_\odot$ Peak and Hierarchical Mergers: The second and third "families" are not new chapters of stellar evolution. They are standard stellar-mass black holes whose gravitational wave signals have traversed varying depths of Phase 3 vacuum viscosity. The extreme mass values—approaching and crossing the pair-instability gap—are exact artifacts of Chirp Mass Inflation. The standard pipeline interprets the causal attenuation of the wave as evidence of an impossibly massive source.

The Spin Degeneracy: Ray et al. attribute the randomized and precessing spins of the heavier subpopulations to chaotic dynamical encounters in globular clusters [6]. RRT proves that this is actually the signature of Cosmic Precession (ω_p). The temporal vector field (T^μ) precesses as a function of redshift. Gravitational waves originating from deeper causal strata suffer phase decoherence, which the Bayesian pipelines mistakenly reconstruct as randomized effective spin (χ_{eff}) and in-plane spin precession (χ_p) of the binary system itself.

As Bergson profoundly warned, the intellect's tendency to spatialize time inevitably leads to insoluble paradoxes [8]. By stripping the universe of its true temporal duration and treating the vacuum as an empty geometric abstraction, standard cosmology is forced to invent invisible entities—be it dark matter, dark energy, or "hierarchical black holes"—to balance its equations.

5 Conclusion

The publication of the GWTC-4 subpopulation analysis by Ray et al. (2026) does not represent the discovery of new astrophysical formation channels; it represents the real-time, empirical validation of the *Referential Relativity Theory*'s Causal Fatigue Test.

The anomalous masses and "three families" reported by the LIGO-Virgo-KAGRA collaborations are precisely the "fantastical populations of impossibly massive black holes" that RRT Volume IV predicted would emerge from standard data processing pipelines in early 2026. The anomaly is not in the stars, but in the ΛCDM metric.

Gravitational wave astronomy is arguably the most sensitive probe of the cosmos ever devised. However, until the international collaborations replace the FLRW metric with the causal hydrodynamic equations of RRT—acknowledging the Phase 3 vacuum impedance (β)—they will continue to catalog algorithmic phantoms. The mathematical framework to correct these distortions is now open and auditable via the RRT Cosmological Engine. The era of dark parameters and ad hoc astrophysical excuses is over.

References

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